Local global hybrid adaptive stabilization via a quadratic Lyapunov function and a transport PDE

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April 27, 2026

We consider next system

$$\dot{x} = f(x) + G(x)\big(u + \Phi(t)\theta^\star\big), \qquad x \in \mathbb{R}^n, \quad u \in \mathbb{R}^m, \quad \theta^\star \in \mathbb{R}^q, \tag{1}$$

where $f \in C^1(\mathbb{R}^n, \mathbb{R}^n)$, $G \in C^1(\mathbb{R}^n, \mathbb{R}^{n\times m})$, $f(0) = 0$, and

$$\Phi \in L^\infty([0,\infty), \mathbb{R}^{m\times q}).$$

is a known regressor. The unknown vector $\theta^\star$ is assumed to be constant.

Let

$$A := Df(0), \qquad B := G(0).$$

Assume that there exists $K \in \mathbb{R}^{m\times n}$ such that

$$A_{\mathrm{cl}} := A + BK$$

is Hurwitz. Choose $P = P^\top > 0$ as the solution of

$$A_{\mathrm{cl}}^\top P + PA_{\mathrm{cl}} = -Q, \qquad Q = Q^\top > 0,$$

and define

$$W(x) := x^\top P x. \tag{2}$$

As in the previous case, there exist constants $a_{\mathrm{loc}} > 0$ and $c_{\mathrm{loc}} > 0$ such that

$$\nabla W(x)^\top \big(f(x) + G(x)Kx\big) \le -c_{\mathrm{loc}} W(x), \qquad 0 < W(x) \le a_{\mathrm{loc}}. \tag{3}$$

The number a_{loc} is the state local Lyapunov radius.

Let $\bar{\Gamma} = \bar{\Gamma}^\top > 0$ be fixed and let

$$\Gamma := \gamma\bar{\Gamma}, \qquad \gamma > 0.$$

For the parameter estimate $\hat{\theta}(t)$, define

$$\tilde{\theta}(t) := \hat{\theta}(t) - \theta^\star$$

and the extended Lyapunov function

$$V_\gamma(x, \tilde{\theta}) := W(x) + \frac{1}{2}\tilde{\theta}^\top \Gamma^{-1}\tilde{\theta} = W(x) + \frac{1}{2\gamma}\tilde{\theta}^\top \bar{\Gamma}^{-1}\tilde{\theta}. \tag{4}$$

Assumption 1. There exists a known compact convex set $\Theta \subset \mathbb{R}^q$ such that

$$\theta^\star \in \Theta.$$

The estimate $\hat{\theta}(t)$ is updated by a projection operator

$$\mathrm{Proj}_\Theta(\hat\theta, y) := \arg\min_{v\in T_\Theta(\hat\theta)} \frac{1}{2}(v-y)^\top \Gamma^{-1}(v-y), \text{ where}$$

$$T_\Theta(\hat\theta) := \mathrm{cl}\left\{\lambda(\theta-\hat\theta) : \lambda \geq 0,\ \theta\in\Theta\right\}.$$

which satisfies

$$\hat\theta(t) \in \Theta \qquad \forall t \geq 0$$

and projection inequality

$$(\hat\theta - \theta^\star)^\top \Gamma^{-1}\left[\mathrm{Proj}_\Theta(\hat\theta, y) - y\right] \leq 0 \tag{5}$$

holds for all $\hat\theta \in \Theta$, $\theta^\star \in \Theta$, and all $y \in \mathbb{R}^q$.

Define

$$C_\Theta := \frac{1}{2} \sup_{\hat\theta,\theta\in\Theta} (\hat\theta - \theta)^\top \bar\Gamma^{-1}(\hat\theta-\theta). \tag{6}$$

Lemma 1. *Under Assumption 1, along every projected trajectory,*

$$\frac{1}{2}\tilde\theta(t)^\top \Gamma^{-1}\tilde\theta(t) \leq \frac{C_\Theta}{\gamma}.$$

Proof. Indeed, $\hat\theta(t)\in\Theta$ and $\theta^\star\in\Theta$. Hence, by the definition of C_Θ,

$$\frac{1}{2}(\hat\theta(t)-\theta^\star)^\top \bar\Gamma^{-1}(\hat\theta(t)-\theta^\star) \leq C_\Theta.$$

Since $\Gamma^{-1} = \gamma^{-1}\bar\Gamma^{-1}$, this gives

$$\frac{1}{2}\tilde\theta(t)^\top \Gamma^{-1}\tilde\theta(t) = \frac{1}{2\gamma}\tilde\theta(t)^\top \bar\Gamma^{-1}\tilde\theta(t) \leq \frac{C_\Theta}{\gamma}.$$

□

The local adaptive law is

$$u_{\mathrm{loc}}(x,\hat\theta,t) := Kx - \Phi(t)\hat\theta, \tag{7}$$

with

$$\dot{\hat\theta} = \mathrm{Proj}_\Theta\Big(\hat\theta,\ \Gamma\Phi(t)^\top G(x)^\top \nabla W(x)\Big). \tag{8}$$

Lemma 2. *Under the local adaptive law* (7)*–*(8)*,*

$$\dot V_\gamma(x,\tilde\theta) \leq -c_{\mathrm{loc}} W(x) \tag{9}$$

whenever $0 < W(x) \leq a_{\mathrm{loc}}$. *Consequently, for every level* $\bar a \in (0, a_{\mathrm{loc}}]$*, the set*

$$\mathcal{E}^\gamma_{\bar a} := \left\{(x,\hat\theta) : V_\gamma(x, \hat\theta - \theta^\star) \leq \bar a\right\} \tag{10}$$

is positively invariant for the local adaptive system, and every trajectory starting in $\mathcal{E}^\gamma_{\bar a}$ *satisfies*

$$x(t) \to 0 \qquad (t \to +\infty).$$

Proof. Under (7),

$$u_{\mathrm{loc}} + \Phi(t)\theta^\star = Kx - \Phi(t)\hat\theta + \Phi(t)\theta^\star = Kx - \Phi(t)\tilde\theta.$$

Hence

$$\dot x = f(x) + G(x)Kx - G(x)\Phi(t)\tilde\theta.$$

Differentiating V_γ, we get

$$\begin{aligned}
\dot V_\gamma &= \nabla W(x)^\top \dot x + \tilde\theta^\top \Gamma^{-1}\dot{\hat\theta} \\
&= \nabla W(x)^\top \big(f(x) + G(x)Kx\big) - \nabla W(x)^\top G(x)\Phi(t)\tilde\theta \\
&\quad + \tilde\theta^\top \Gamma^{-1} \operatorname{Proj}_\Theta\Big(\hat\theta,\, \Gamma\Phi(t)^\top G(x)^\top \nabla W(x)\Big).
\end{aligned}$$

Add and subtract $\Gamma\Phi(t)^\top G(x)^\top \nabla W(x)$ inside the projection term. Then

$$\begin{aligned}
\dot V_\gamma = {}& \nabla W(x)^\top \big(f(x) + G(x)Kx\big) \\
& - \nabla W(x)^\top G(x)\Phi(t)\tilde\theta + \tilde\theta^\top \Phi(t)^\top G(x)^\top \nabla W(x) \\
& + \tilde\theta^\top \Gamma^{-1}\Big[\operatorname{Proj}_\Theta\Big(\hat\theta,\, \Gamma\Phi(t)^\top G(x)^\top \nabla W(x)\Big) - \Gamma\Phi(t)^\top G(x)^\top \nabla W(x)\Big].
\end{aligned}$$

The second and third terms cancel. By the projection inequality (5), the last term is nonpositive. Therefore

$$\dot V_\gamma \le \nabla W(x)^\top \big(f(x) + G(x)Kx\big).$$

Using (3), we obtain

$$\dot V_\gamma \le -c_{\text{loc}} W(x), \qquad 0 < W(x) \le a_{\text{loc}}.$$

Now let $(x(0), \hat\theta(0)) \in \mathcal{E}_{\bar a}^{\gamma}$ with $\bar a \le a_{\text{loc}}$. Since

$$W(x) \le V_\gamma(x, \tilde\theta),$$

we have

$$V_\gamma(x(0), \tilde\theta(0)) \le \bar a \quad \Longrightarrow \quad W(x(0)) \le \bar a \le a_{\text{loc}}.$$

As long as the trajectory remains in $\mathcal{E}_{\bar a}^{\gamma}$, one has $W(x(t)) \le \bar a \le a_{\text{loc}}$, and hence

$$\dot V_\gamma \le -c_{\text{loc}} W \le 0.$$

Thus V_γ cannot increase past $\bar a$, and $\mathcal{E}_{\bar a}^{\gamma}$ is positively invariant.

Furthermore,

$$c_{\text{loc}} \int_0^\infty W(x(t))\, dt \le V_\gamma(x(0), \tilde\theta(0)).$$

Since $P > 0$, this implies $x \in L^2([0, \infty))$. The state $x(t)$ is bounded because $W(x(t)) \le \bar a$, and $\hat\theta(t) \in \Theta$ by projection. Hence $\dot x(t)$ is bounded on $[0, \infty)$, so $x(t)$ is uniformly continuous. By Barbalat's lemma,

$$x(t) \to 0.$$

□

Assumption 2.

$$G(x)^\top \nabla W(x) \ne 0 \qquad \text{for all } x \ne 0. \tag{11}$$

Define the outer adaptive law by

$$u_{\text{ext}}(x, \hat\theta, t) := -\Phi(t)\hat\theta - \frac{W(x) + \nabla W(x)^\top f(x)}{\|G(x)^\top \nabla W(x)\|^2} G(x)^\top \nabla W(x), \qquad x \ne 0, \tag{12}$$

with the same projected update

$$\dot{\hat\theta} = \operatorname{Proj}_\Theta\Big(\hat\theta,\, \Gamma\Phi(t)^\top G(x)^\top \nabla W(x)\Big). \tag{13}$$

Lemma 3. *Under the outer adaptive law* (12)–(13),

$$\dot{V}_\gamma(x, \tilde{\theta}) \le -W(x) \tag{14}$$

whenever $x \neq 0$.

Proof. Under (12),

$$u_{\text{ext}} + \Phi(t)\theta^\star = -\Phi(t)\tilde{\theta} - \frac{W(x) + \nabla W(x)^\top f(x)}{\|G(x)^\top \nabla W(x)\|^2} G(x)^\top \nabla W(x).$$

Therefore,

$$\begin{aligned} \dot{x} &= f(x) - G(x)\Phi(t)\tilde{\theta} \\ &\quad - G(x)\frac{W(x) + \nabla W(x)^\top f(x)}{\|G(x)^\top \nabla W(x)\|^2} G(x)^\top \nabla W(x). \end{aligned}$$

Differentiating V_γ, using the projected update, and adding and subtracting the unprojected update exactly as in the proof of Lemma 2, we obtain

$$\begin{aligned} \dot{V}_\gamma &\le \nabla W(x)^\top f(x) \\ &\quad - \frac{W(x) + \nabla W(x)^\top f(x)}{\|G(x)^\top \nabla W(x)\|^2} \nabla W(x)^\top G(x) G(x)^\top \nabla W(x). \end{aligned}$$

But

$$\nabla W(x)^\top G(x) G(x)^\top \nabla W(x) = \left\| G(x)^\top \nabla W(x) \right\|^2.$$

Thus

$$\dot{V}_\gamma \le \nabla W(x)^\top f(x) - \big(W(x) + \nabla W(x)^\top f(x)\big) = -W(x).$$

□

Lemma 4. *Assume Assumption 1 and Assumption 2. Let* $\bar{a} > 0$ *and choose* $\gamma > 0$ *such that*

$$\frac{C_\Theta}{\gamma} < \bar{a}.$$

Define

$$r_\gamma := \bar{a} - \frac{C_\Theta}{\gamma} > 0. \tag{15}$$

Consider the outer adaptive law (12)–(13) *as long as*

$$W(x) > r_\gamma.$$

Then the trajectory reaches the implementable switching set

$$\Omega_\gamma := \{x \in \mathbb{R}^n : W(x) \le r_\gamma\} \tag{16}$$

in finite time. Moreover, at the first entrance time

$$\tau_\gamma := \inf\{t \ge 0 : W(x(t)) \le r_\gamma\},$$

one has

$$V_\gamma(x(\tau_\gamma), \tilde{\theta}(\tau_\gamma)) \le \bar{a}.$$

Consequently, switching at time τ_γ *to the local adaptive law places the closed-loop trajectory inside*

$$\mathcal{E}_{\bar{a}}^\gamma = \{(x, \hat{\theta}) : V_\gamma(x, \hat{\theta} - \theta^\star) \le \bar{a}\}.$$

Proof. By the projection property and the definition of C_Θ, along every projected trajectory,

$$\frac{1}{2}\tilde{\theta}(t)^\top \Gamma^{-1}\tilde{\theta}(t) \le \frac{C_\Theta}{\gamma}.$$

Therefore, for all $t \ge 0$,

$$V_\gamma(x(t),\tilde{\theta}(t)) = W(x(t)) + \frac{1}{2}\tilde{\theta}(t)^\top \Gamma^{-1}\tilde{\theta}(t) \le W(x(t)) + \frac{C_\Theta}{\gamma}. \tag{17}$$

If

$$W(x(0)) \le r_\gamma,$$

then $\tau_\gamma = 0$, and by (17),

$$V_\gamma(x(0),\tilde{\theta}(0)) \le r_\gamma + \frac{C_\Theta}{\gamma} = \bar{a}.$$

Thus the claim is immediate.

Assume now that

$$W(x(0)) > r_\gamma.$$

Then $x(0) \ne 0$, because $r_\gamma > 0$. As long as

$$W(x(t)) > r_\gamma,$$

we also have $x(t) \ne 0$. Hence the outer adaptive law is well defined. By Lemma 3,

$$\dot{V}_\gamma(x(t),\tilde{\theta}(t)) \le -W(x(t)) < -r_\gamma.$$

We prove finite time entrance by contradiction. Suppose that the trajectory never reaches Ω_γ. Then

$$W(x(t)) > r_\gamma \qquad \forall t \ge 0.$$

Consequently,

$$\dot{V}_\gamma(x(t),\tilde{\theta}(t)) \le -r_\gamma \qquad \forall t \ge 0.$$

Integrating, we get

$$V_\gamma(x(t),\tilde{\theta}(t)) \le V_\gamma(x(0),\tilde{\theta}(0)) - r_\gamma t.$$

The right-hand side becomes negative for sufficiently large t, whereas

$$V_\gamma(x(t),\tilde{\theta}(t)) \ge 0$$

by definition. This contradiction proves that $\tau_\gamma < \infty$.

At $t = \tau_\gamma$, by definition of the first entrance time,

$$W(x(\tau_\gamma)) \le r_\gamma.$$

Using again the parameter bound,

$$\begin{aligned} V_\gamma(x(\tau_\gamma),\tilde{\theta}(\tau_\gamma)) &= W(x(\tau_\gamma)) + \frac{1}{2}\tilde{\theta}(\tau_\gamma)^\top \Gamma^{-1}\tilde{\theta}(\tau_\gamma) \\ &\le r_\gamma + \frac{C_\Theta}{\gamma} \\ &= \bar{a}. \end{aligned}$$

Therefore

$$(x(\tau_\gamma),\hat{\theta}(\tau_\gamma)) \in \mathcal{E}^\gamma_{\bar{a}}.$$

□

Theorem 1. *Assume:*

1. *$A + BK$ is Hurwitz;*
2. *local estimate* (3) *holds on* $0 < W \le a_{\mathrm{loc}}$*;*
3. *Assumption 1 holds;*
4. *Assumption 2 holds.*

Then for every desired Lyapunov margin

$$0 < \eta < a_{\mathrm{loc}},$$

there exist a local Lyapunov level $\bar{a}$*, an implementable switching level* r_γ*, and a gain threshold* $\gamma_\eta > 0$ *such that for every* $\gamma \ge \gamma_\eta$*, the following hybrid adaptive controller is well defined:*

$$\textit{use } u_{\mathrm{ext}} \textit{ while } W(x) > r_\gamma, \qquad \textit{switch to } u_{\mathrm{loc}} \textit{ once } W(x) \le r_\gamma.$$

The outer adaptive trajectory reaches the switching set

$$\Omega_\gamma = \{x : W(x) \le r_\gamma\}$$

in finite time. After the switch, the trajectory belongs to

$$\mathcal{E}^\gamma_{\bar{a}} = \{(x, \hat{\theta}) : V_\gamma(x, \hat{\theta} - \theta^\star) \le \bar{a}\},$$

remains in this set for all future time under the local adaptive law, and satisfies

$$x(t) \to 0 \qquad (t \to +\infty).$$

More precisely, one may choose

$$\bar{a} := a_{\mathrm{loc}} - \eta, \tag{18}$$

and any

$$\gamma_\eta > \frac{C_\Theta}{\bar{a}}. \tag{19}$$

For every $\gamma \ge \gamma_\eta$*, define*

$$r_\gamma := \bar{a} - \frac{C_\Theta}{\gamma} > 0. \tag{20}$$

Proof. Fix $\eta \in (0, a_{\mathrm{loc}})$, and define

$$\bar{a} := a_{\mathrm{loc}} - \eta.$$

Then

$$0 < \bar{a} < a_{\mathrm{loc}}.$$

Hence

$$V_\gamma(x, \tilde{\theta}) \le \bar{a} \quad \Longrightarrow \quad W(x) \le \bar{a} < a_{\mathrm{loc}}.$$

Thus, once the trajectory reaches $\mathcal{E}^\gamma_{\bar{a}}$, the state lies strictly inside the local Lyapunov region where Lemma 2 applies.

Choose γ_η so that

$$\gamma_\eta > \frac{C_\Theta}{\bar{a}}.$$

Then for every $\gamma \ge \gamma_\eta$,

$$\frac{C_\Theta}{\gamma} < \bar{a}.$$

Therefore the number

$$r_\gamma := \bar{a} - \frac{C_\Theta}{\gamma}$$

is strictly positive.

We use the following hybrid rule. If

$$W(x(0)) \le r_\gamma,$$

then the local adaptive law is used immediately. In this case, by the parameter bound,

$$\begin{aligned} V_\gamma(x(0), \tilde{\theta}(0)) &= W(x(0)) + \frac{1}{2}\tilde{\theta}(0)^\top \Gamma^{-1} \tilde{\theta}(0) \\ &\le r_\gamma + \frac{C_\Theta}{\gamma} \\ &= \bar{a}. \end{aligned}$$

Thus

$$(x(0), \hat{\theta}(0)) \in \mathcal{E}^\gamma_{\bar{a}}.$$

Lemma 2 then implies positive invariance of $\mathcal{E}^\gamma_{\bar{a}}$ and convergence $x(t) \to 0$.

Now suppose that

$$W(x(0)) > r_\gamma.$$

Then the outer adaptive law is used as long as

$$W(x) > r_\gamma.$$

By Lemma 4, the trajectory reaches

$$\Omega_\gamma = \{x : W(x) \le r_\gamma\}$$

in finite time. Let

$$\tau_\gamma := \inf\{t \ge 0 : W(x(t)) \le r_\gamma\}$$

be the first entrance time. Lemma 4 gives

$$V_\gamma(x(\tau_\gamma), \tilde{\theta}(\tau_\gamma)) \le \bar{a}.$$

Therefore

$$(x(\tau_\gamma), \hat{\theta}(\tau_\gamma)) \in \mathcal{E}^\gamma_{\bar{a}}.$$

At time τ_γ, switch to the local adaptive law. Since

$$\bar{a} < a_{\mathrm{loc}},$$

Lemma 2 implies that $\mathcal{E}^\gamma_{\bar{a}}$ is positively invariant under the local adaptive law. Consequently,

$$V_\gamma(x(t), \tilde{\theta}(t)) \le \bar{a} \qquad \forall t \ge \tau_\gamma,$$

and therefore

$$W(x(t)) \le \bar{a} < a_{\mathrm{loc}} \qquad \forall t \ge \tau_\gamma.$$

Again by Lemma 2,

$$x(t) \to 0 \qquad (t \to +\infty).$$

□

Remark 1. *The theorem shows the precise role of the adaptive gain. The compact parameter set produces the parameter bound*

$$\frac{1}{2}\tilde{\theta}^\top \Gamma^{-1}\tilde{\theta} \le \frac{C_\Theta}{\gamma}.$$

By increasing γ, this bound can be made smaller than any prescribed local level $\bar{a} > 0$. The implementable switching level is then chosen as

$$r_\gamma = \bar{a} - \frac{C_\Theta}{\gamma} > 0.$$

Thus

$$W(x) \le r_\gamma \quad \Longrightarrow \quad V_\gamma(x, \tilde{\theta}) \le \bar{a},$$

even though V_γ itself depends on the unknown parameter $\theta^\star$. The controller therefore switches using only the measurable quantity $W(x)$, while the proof uses the extended Lyapunov function to certify entrance into the local invariant set.

1 Example: adaptive roll stabilization of a ship

We now illustrate the proposed hybrid adaptive design on a nonlinear roll model of a ship. The state is

$$x = \begin{bmatrix} \theta \\ p \end{bmatrix} \in \mathbb{R}^2,$$

where θ is the roll angle and $p = \dot{\theta}$ is the roll rate. The control input is a scalar torque $u \in \mathbb{R}$. The dynamics are

$$\dot{\theta} = p, \qquad \dot{p} = -cp - k\sin\theta + u + d(t), \tag{21}$$

where $c > 0$ is the damping coefficient, $k > 0$ is the restoring coefficient, and $d(t)$ is an unknown matched wave induced disturbance.

Equivalently,

$$\dot{x} = f(x) + B\big(u + d(t)\big), \tag{22}$$

where

$$f(x) = \begin{bmatrix} p \\ -cp - k\sin\theta \end{bmatrix}, \qquad B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

The parameters are chosen as

$$c = 0.85, \qquad k = 2.20.$$

The disturbance is assumed to have the matched parametric form

$$d(t) = \varphi(t)^\top \theta^\star, \tag{23}$$

where $\theta^\star \in \mathbb{R}^q$ is an unknown constant vector and $\varphi(t) \in \mathbb{R}^q$ is a known bounded regressor. We use a harmonic regressor

$$\varphi(t) = \begin{bmatrix} \sin(\omega_1 t) \\ \cos(\omega_1 t) \\ \vdots \\ \sin(\omega_N t) \\ \cos(\omega_N t) \end{bmatrix}, \tag{24}$$

with frequencies

$$\omega_i \in [0.35, 3.50], \qquad i = 1, \dots, N.$$

The compact parameter set is chosen as the Euclidean ball

$$\Theta = \{\theta \in \mathbb{R}^q : \|\theta\| \le R_\Theta\}, \qquad R_\Theta = 0.65. \tag{25}$$

We take

$$\bar{\Gamma} = I_q, \qquad \Gamma = \gamma I_q.$$

Since $\hat{\theta}(t) \in \Theta$ and $\theta^\star \in \Theta$, one has

$$\|\hat{\theta}(t) - \theta^\star\| \leq 2R_\Theta.$$

Hence

$$C_\Theta = 2R_\Theta^2. \tag{26}$$

For $R_\Theta = 0.65$, this gives

$$C_\Theta = 2(0.65)^2 = 0.845.$$

The linearization of (22) at the origin is

$$A = \begin{bmatrix} 0 & 1 \\ -k & -c \end{bmatrix}, \qquad B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

For the chosen numerical parameters,

$$A = \begin{bmatrix} 0 & 1 \\ -2.20 & -0.85 \end{bmatrix}.$$

We choose the local feedback

$$K = \begin{bmatrix} -3.8 & -2.4 \end{bmatrix}. \tag{27}$$

Then

$$A_{\mathrm{cl}} = A + BK = \begin{bmatrix} 0 & 1 \\ -6.0 & -3.25 \end{bmatrix}.$$

Its eigenvalues are

$$\lambda_{1,2} = -1.625 \pm 1.8329\, i.$$

Therefore A_{cl} is Hurwitz.

Let $P = P^\top > 0$ be the solution of the Lyapunov equation

$$A_{\mathrm{cl}}^\top P + P A_{\mathrm{cl}} = -I. \tag{28}$$

Numerically,

$$P = \begin{bmatrix} 1.34775641 & 0.08333333 \\ 0.08333333 & 0.17948718 \end{bmatrix}. \tag{29}$$

The state Lyapunov function is

$$W(x) = x^\top P x. \tag{30}$$

Since A_{cl} is Hurwitz there exists $a_{\mathrm{loc}} > 0$ such that the nominal local decay estimate

$$\nabla W(x)^\top \big(f(x) + BKx\big) \leq -c_{\mathrm{loc}} W(x)$$

holds for $0 < W(x) \leq a_{\mathrm{loc}}$. In the example we take

$$a_{\mathrm{loc}} = 0.20.$$

The local adaptive controller is

$$u_{\mathrm{loc}}(x, \hat{\theta}, t) = Kx - \varphi(t)^\top \hat{\theta}. \tag{31}$$

The projected adaptive law is

$$\dot{\hat{\theta}} = \mathrm{Proj}_\Theta\left(\hat{\theta},\, \Gamma \varphi(t) B^\top \nabla W(x)\right). \tag{32}$$

Since

$$\nabla W(x) = 2Px,$$

we have

$$B^\top \nabla W(x) = 2B^\top Px.$$

Thus, in explicit form,

$$\dot{\hat{\theta}} = \mathrm{Proj}_\Theta\left(\hat{\theta},\, 2\gamma\,\varphi(t)B^\top Px\right). \tag{33}$$

The corresponding extended Lyapunov function is

$$V_\gamma(x,\tilde{\theta}) = x^\top Px + \frac{1}{2\gamma}\|\tilde{\theta}\|^2, \qquad \tilde{\theta} = \hat{\theta} - \theta^\star. \tag{34}$$

By the general local adaptive result, for all $0 < W(x) \le a_{\mathrm{loc}}$,

$$\dot{V}_\gamma(x,\tilde{\theta}) \le -c_{\mathrm{loc}} W(x).$$

Therefore every sublevel set

$$\mathcal{E}^\gamma_{\bar{a}} = \left\{(x,\hat{\theta}) : V_\gamma(x,\hat{\theta}-\theta^\star) \le \bar{a}\right\}, \qquad 0 < \bar{a} < a_{\mathrm{loc}},$$

is positively invariant under the local adaptive closed-loop system, and

$$x(t) \to 0 \qquad (t \to +\infty).$$

In the outer region, the adaptive controller is chosen according to the transport construction. Since the input is scalar, define

$$g(x) := B^\top \nabla W(x) = 2B^\top Px. \tag{35}$$

The theoretical outer law is

$$u_{\mathrm{ext}}(x,\hat{\theta},t) = -\varphi(t)^\top\hat{\theta} - \frac{W(x) + \nabla W(x)^\top f(x)}{g(x)}, \tag{36}$$

whenever $g(x) \neq 0$. Equivalently, this is the scalar-input specialization of

$$u_{\mathrm{ext}} = -\varphi(t)^\top\hat{\theta} - \frac{W(x) + \nabla W(x)^\top f(x)}{\|B^\top \nabla W(x)\|^2} B^\top \nabla W(x).$$

The adaptive update in the outer mode is

$$\dot{\hat{\theta}} = \mathrm{Proj}_\Theta\left(\hat{\theta},\, \Gamma\varphi(t)B^\top \nabla W(x)\right). \tag{37}$$

Thus the same projected estimator is used in both modes.

Substituting (36) into the dynamics gives

$$\dot{x} = f(x) - B\varphi(t)^\top\tilde{\theta} - B\frac{W(x) + \nabla W(x)^\top f(x)}{B^\top \nabla W(x)}.$$

Using the adaptive update and the projection inequality, one obtains

$$\dot{V}_\gamma(x,\tilde{\theta}) \le \nabla W(x)^\top f(x) - \big(W(x) + \nabla W(x)^\top f(x)\big) = -W(x),$$

provided $B^\top \nabla W(x) \neq 0$. Hence, as long as the outer law is well defined,

$$\dot{V}_\gamma(x,\tilde{\theta}) \le -W(x). \tag{38}$$

For the present single-input two-dimensional example,

$$B^\top \nabla W(x) = 2 \begin{bmatrix} 0 & 1 \end{bmatrix} P \begin{bmatrix} \theta \\ p \end{bmatrix} = 2(P_{21}\theta + P_{22}p).$$

Therefore the nondegeneracy condition is not global in $\mathbb{R}^2$, since $B^\top \nabla W(x)$ may vanish on a line through the origin. The rigorous application of the theorem is therefore understood on any outer region, or along any outer trajectory, where

$$|B^\top \nabla W(x)| > 0.$$

We choose

$$\eta = 0.06, \qquad a_{\mathrm{loc}} = 0.20.$$

Therefore

$$\bar{a} = a_{\mathrm{loc}} - \eta = 0.14. \tag{39}$$

Since

$$C_\Theta = 0.845,$$

the gain threshold is chosen as

$$\gamma_\eta > \frac{C_\Theta}{\bar{a}} = \frac{0.845}{0.14} \approx 6.0357. \tag{40}$$

In the simulation,

$$\gamma_\eta = 6.3375, \qquad \gamma = 7.921875.$$

Hence

$$\frac{C_\Theta}{\gamma} = \frac{0.845}{7.921875} = 0.106666\ldots < \bar{a}.$$

The implementable switching level is therefore

$$r_\gamma = \bar{a} - \frac{C_\Theta}{\gamma} = 0.14 - 0.106666\ldots = 0.033333\ldots. \tag{41}$$

The hybrid adaptive controller is

$$u(t) = \begin{cases} u_{\mathrm{ext}}(x(t), \hat{\theta}(t), t), & W(x(t)) > r_\gamma, \\ u_{\mathrm{loc}}(x(t), \hat{\theta}(t), t), & W(x(t)) \le r_\gamma. \end{cases} \tag{42}$$

The switch is performed only once. After the first entrance into the set

$$\Omega_\gamma = \{x \in \mathbb{R}^2 : W(x) \le r_\gamma\},$$

the controller remains in the local adaptive mode.